TUTORIAL – 1: SDLC & STLC PHASES

Aim

To understand the **Software Development Life Cycle (SDLC)** and **Software Testing Life Cycle (STLC)** phases for a real-world software project and identify the testing activities performed during each stage of software development.

Description

The **Software Development Life Cycle (SDLC)** is a structured process used to develop high-quality software. It defines all activities from requirement gathering to software maintenance.

The **Software Testing Life Cycle (STLC)** is a sequence of testing activities performed to ensure that the developed software meets user requirements and is free from defects.

Both SDLC and STLC work together to produce reliable and high-quality software.

SDLC Phases

1. **Requirement Analysis**
    - Gather and analyze user requirements.
    - Prepare the Software Requirement Specification (SRS).
2. **System Design**
    - Design system architecture, database, and user interface.
3. **Implementation (Coding)**
    - Developers write the source code.
4. **Testing**
    - Testers verify that the software works according to the requirements.
5. **Deployment**
    - Software is released to users.
6. **Maintenance**
    - Fix bugs, improve features, and provide updates after deployment.

STLC Phases

1. Requirement Analysis
2. Test Planning
3. Test Case Development
4. Test Environment Setup

5. Test Execution
6. Defect Reporting
7. Test Closure

Problem Statement

Study the **Student Management System** and identify the SDLC and STLC phases involved in its development. Perform basic testing on the Login Module, prepare a sample test case, execute it manually, and document the testing activities.

How to Perform This Practical (Student Procedure)

Step 1: Study the Software Project

Consider the **Student Management System** with the following modules:

- Login
- Student Registration
- Student Details
- Search Student
- Logout

Step 2: Identify SDLC Phases

Write the activities performed in each SDLC phase.

SDLC Phase	Activity
Requirement Analysis	Collect user requirements
Design	Design UI and Database
Coding	Develop Login Module
Testing	Verify Login functionality
Deployment	Install software
Maintenance	Fix bugs and update software

Step 3: Identify STLC Phases
Map the testing activities.

STLC Phase	Activity
Requirement Analysis	Review SRS
Test Planning	Prepare Test Plan
Test Case Development	Write Test Cases
Test Environment Setup	Configure system
Test Execution	Execute test cases
Defect Reporting	Report bugs
Test Closure	Prepare Test Summary

Step 4: Execute Sample Test Case
Use the following input:
Username: admin
Password: admin123
Expected Result:
Login Successful

Step 5: Record the Results
Compare:
Expected Result
vs
Actual Result
Mark:
PASS or FAIL

Step 6: Prepare Observation
Example:
✔ Login works correctly.
✔ No defects found.
✔ Test case passed successfully.

Implementation Details
The **Student Management System** is developed using the **Waterfall SDLC Model**. The Login Module is tested manually by executing test cases using valid credentials and comparing the expected results with the actual output.

Parameter Details

Parameter	Details
Project Name	Student Management System
Application/Software	Student Management System
Module Name	Login Module
SDLC Model	Waterfall Model
Programming Language	Python
IDE/Tool Used	PyCharm / Visual Studio Code
Input/Test Data	Username: admin Password: admin123

Source Code
```
username = input("Enter Username: ")
password = input("Enter Password: ")

if username == "admin" and password == "admin123":
    print("Login Successful")
else:
    print("Invalid Username or Password")
```

Test Case Detailed

Field	Details
Test Case ID	TC001
Requirement ID	R001
Module Name	Login Module
Feature Under Test	User Authentication
Test Scenario	Login using valid username and password
Test Objective	Verify successful login
Test Type (Unit/Integration/System/UAT)	System Testing
Test Technique (Black Box/White Box)	Black Box Testing
Priority	High
Severity	Critical
Preconditions	Application is running and Login page is displayed
Postconditions	User enters Home Page
Test Environment	Windows 10
Browser/OS (if applicable)	Google Chrome / Windows 10
Test Data	admin / admin123
Expected Result	Login Successful
Actual Result	Login Successful

Status (Pass/Fail)	Pass
Executed By	Student Name
Execution Date	DD/MM/YYYY

Test Case Execution

Step No.	Test Steps	Input	Expected Result	Actual Result	Status
1	Open Application	-	Login page displayed	Login page displayed	Pass
2	Enter Username	admin	Username accepted	Username accepted	Pass
3	Enter Password	admin123	Password accepted	Password accepted	Pass
4	Click Login Button	Click	Login Successful	Login Successful	Pass

Testing Activities Performed

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1	Requirement Analysis	Requirement Analysis	Analyze software requirements (SRS)	Software Requirement Specification (SRS)
2	System Design	Test Planning	Prepare test plan and testing strategy	Test Plan
3	Implementation (Coding)	Test Case Development	Design test cases for Login Module	Test Case Document

4	Testing	Test Environment Setup	Configure testing environment	Test Environment Setup Report
5	Testing	Test Execution	Execute test cases and record results	Test Execution Report
6	Deployment & Maintenance	Test Closure	Prepare final testing summary and close testing	Test Summary Report

Observation

Sr. No.	Observation	Remarks
1	SDLC phases identified correctly	Completed
2	STLC phases mapped successfully	Completed
3	Login module tested successfully	Pass
4	Expected and Actual Results matched	Pass
5	No defects found during execution	Completed

Output

Student Management System

Username : admin
Password : admin123

Login Successful

Test Case Execution Status : PASS

Testing Completed Successfully

Program Outcome

- Understood the phases of **Software Development Life Cycle (SDLC)** and **Software Testing Life Cycle (STLC)**.
- Learned how testing activities are mapped to each SDLC phase.
- Executed a sample test case for the Login Module.
- Compared expected and actual results to determine the test status.
- Gained practical knowledge of the software development and testing process.

Solve the Below Mention Questionnaires

1. What does SDLC stand for?

A) Software Design Life Cycle

B) Software Development Life Cycle

C) System Design Life Cycle

D) Software Deployment Life Cycle

Answer: B) Software Development Life Cycle

2. Which is the first phase of SDLC?

A) Testing

B) Deployment

C) Requirement Analysis

D) Maintenance

Answer: C) Requirement Analysis

3. What does STLC stand for?
A) Software Technology Life Cycle
B) Software Testing Life Cycle
C) System Testing Life Cycle
D) Software Test Logic Cycle
Answer: B) Software Testing Life Cycle

4. Which STLC phase involves preparing test cases?
A) Requirement Analysis
B) Test Planning
C) Test Case Development
D) Test Closure
Answer: C) Test Case Development
5. Which testing is performed by developers during coding?
A) System Testing
B) Integration Testing
C) Unit Testing
D) Acceptance Testing
Answer: C) Unit Testing

Conclusion / Outcome

- Understood the purpose and phases of SDLC and STLC.
- Identified the relationship between software development and software testing activities.
- Performed manual testing of the Login Module using a sample test case.
- Successfully recorded the execution results and verified that the application met the expected requirements.
- Learned how SDLC and STLC work together to ensure the development of reliable, high-quality software.